

Design of Real-time and Low-cost Electronic Stethoscope

Trang Nguyen Minh, Hanh Nguyen Hong, Huy-Dung Han, Loan Pham-Nguyen*

School of Electrical and Electronic Engineering, Hanoi University of Science and Technology
Hanoi, Vietnam

*loan.phamnguyenthanh@hust.edu.vn

Abstract—Stethoscopes are a vital diagnostic tool in the medical field. The amplification and recording capabilities of a traditional stethoscope are constrained. Electronic stethoscopes have been created to address this issue. Nowadays, doctors frequently use conventional stethoscopes since they are less expensive than electronic ones. In this study, a low-cost electronic stethoscope with a hearing output jack and real-time cardiac sound realization based on the Arduino UNO R3 is presented. To enable the connection between the stethoscope and the laptop, a Bluetooth module was used. With Excel, the heart sound signal can be stored and displayed. It also activates additional processing to determine heart rate.

Index Terms—electronic stethoscope, heart sound, PCG, auscultation, Arduino

I. INTRODUCTION

In 2019, according to the World Health Organization (WHO), cardiovascular diseases (CVDs) were the main root of death, with 17.9 million cases globally (32% of total fatalities). In addition, about 75% of CVD-related deaths are observed in countries with low and moderate incomes [1]. For cardiovascular disease, early detection is essential in starting treatment with counseling and medical care. One of the fundamental techniques to diagnose heart disease is employing a stethoscope to listen to the heartbeat, as known as auscultation.

Understanding the features of the heart sound, known as heart auscultation, is valuable data in diagnosing and monitoring heart conditions such as heart murmurs, valve defects, and irregular heartbeats. PCG stands for phonocardiography, a non-invasive technique to detect and record the sounds produced by the heart during each cardiac cycle [2].

Mechanical (Acoustic) Stethoscopes are the traditional stethoscopes that have been used for decades. Conventional stethoscopes have limitations in amplification and recording capabilities. In auscultation learning, the instructor and medical student can not share the sound. It can not be applied in remote healthcare monitoring and telemedicine consultations. An electronic stethoscope is needed to cope with these issues. Furthermore, a study demonstrated the effectiveness of the digital stethoscope as a tool for teaching auscultatory skills when compared to traditional methods [3].

Air-conduction and contact-conduction electronic stethoscopes are the two popular electronic stethoscopes, depending

on the sensor they employ. The electromagnetic coil or electret capacitor is used as the sensor in the air-conduction electronic stethoscope to gather sound signals with great stability and strong reliability but limited sensitivity. Piezoelectric materials, which have better anti-interference ability and sensitivity, are primarily used as sound sensors in contact-conduction electronic stethoscopes. Unfortunately, any bending or deformation to the piezoelectric components during the manufacturing process reduces the stethoscope's sensitivity [4].

Muhammad et al in [5] demonstrated an Arduino DUE-based digital stethoscope with logging capabilities for post-processing in Matlab and the ability to visualize heart sounds in real-time through a graphic display. In the study on the application of an electronic stethoscope, Oktivasari et al in [6] designed a cost-effective and simple electronic stethoscope to capture and measure real-time heart rate data (unit: BPM). The results of the study show that this system does not show a significant deviation from the conventional heart rate measurement system, only about 2-5%, with a delay of 1.7 - 2 seconds.

In this study, we develop a low-cost electronic stethoscope with heart sound visualization in a smart device via Bluetooth and a 3.5mm hearing output for earphone connection.

II. METHODOLOGY

The block diagram of a proposed electronic stethoscope is presented in Fig. 1. The stethoscope consists of three main blocks: signal conditioning, analog front end (AFE), and signal processing.

A. Signal Conditioning

The signal conditioning part contains a chest piece of the conventional stethoscope to detect the mechanical sound wave (sound of heartbeat) and a high-sensitive microphone attached at the end of the chest piece.

The connected stem, diaphragm, and bell make up the stethoscope's chest piece. The chest piece's smaller round end is known as the bell half. One side of the gadget concentrates on a smaller spectrum of lower-frequency sounds. The large disk-shaped end of the chest piece is called the diaphragm, it is a sealed membrane. This side of the device picks up a higher-frequency sound than the bell half. The sound of the

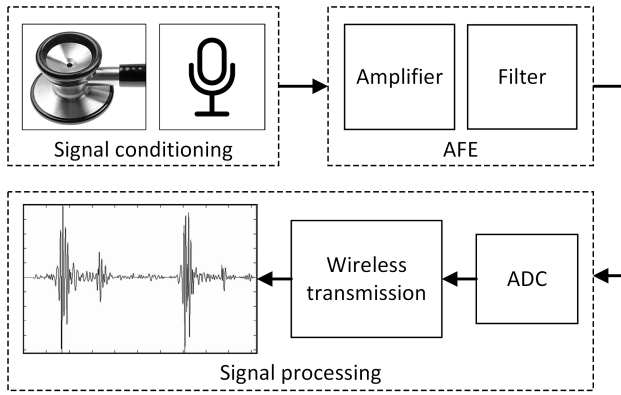


Fig. 1. Block diagram of stethoscope system.

heart will cause the diaphragm, which moves the stethoscope's inner air molecules, to vibrate when a chest piece is placed on a human being's chest. This movement leads to a change in air pressure, which vibrates the membrane of the microphone.

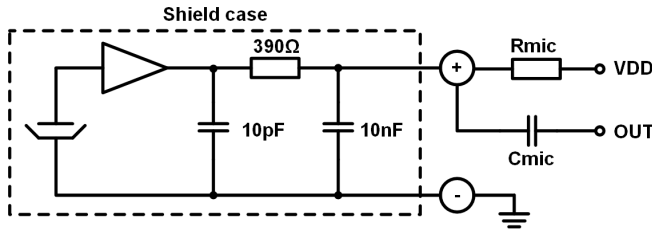


Fig. 2. Microphone biasing circuit.

The microphone functions as a transducer, which is made up of parts for detection and transduction and changes one kind of energy into another. The sensor component produces matching outputs in response to bodily movements. Its output may take the form of strain, displacement, or other quantities. The transduction element is connected to the sensing element's output. As a result, this component was specifically created to react to the sensing element's reaction and provide an electrical signal with a proportional output. The selected microphone, shown in Fig. 2, has a sensitivity of $-45dB$, an output impedance of $2.2k\Omega$, and a frequency range of $20Hz-20kHz$. The microphone operates with a voltage range of $1-10VDC$.

B. Analog-front-end (AFE)

The electrical signal converted by the microphone is then amplified by the analog front end (AFE), consisting of two stages. The gain of the first stage is configured as a fixed-gain voltage. The capacitor $C_{mic} = 10\mu F$ of the microphone removes the DC level of the signal.(Fig. 3).

Before entering the second buffer stage, the signal passes through capacitor C_2 to remove the DC offset from the first stage. In the second stage, the volume of the sound can be adjusted easily by tuning variable resistor R_4 . After being amplified at the two stages, the signal goes through a passive low-pass filter RC to filter out noise (at a frequency higher than

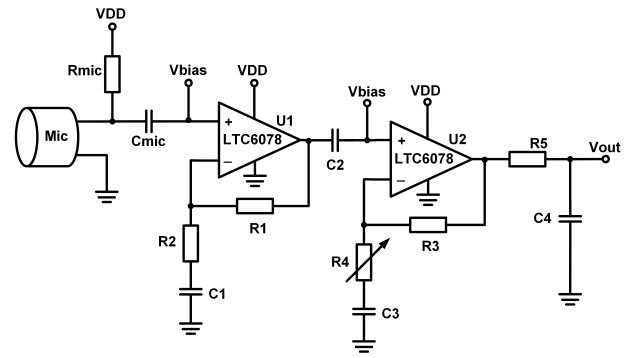


Fig. 3. Amplifier topology.

$200Hz$). For each of these stages, an OpAmp in the LTC6078 package is used. The op-am, powered by a $3.7V$ battery, has a dynamic range of signal from $0V$ to $3.7V$. As the op-am requires a $1.8V$ bias for a high output swing, a voltage divider is used to create a V_{bias} to supply for the positive terminal of the op-amp.

The total gain of the AFE is calculated as:

$$A_v = \frac{R_1}{R_1 + R_2} \times \frac{R_3}{R_3 + R_4} \quad (1)$$

The heart sound has a frequency range from 20 to 200 Hz, so a low pass filter is proposed at the tail of the second stage. The cut-off frequency is determined by:

$$f_L = \frac{1}{2\pi R_5 C_4} \quad (2)$$

C. Signal processing

1) *PCG signal study*: A sample PCG signal is shown in Fig. 4. Systole and diastole are the two phases in a heart cycle. Diastole is the phase when the heart relaxes, while systole is its contracting phase.

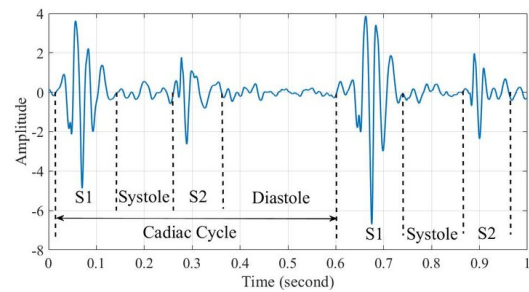


Fig. 4. Sample PCG signal [7].

Systole occurs when the heart valves close and blood is discharged from the ventricles into the pulmonary and systemic circulation. This produces the first heart sound, often known as S1. Low-frequency S1 is most audible at the top of the heart. The heart valves open to let blood flow into the heart as the heart chambers fill with blood during diastole. The second heart sound, sometimes referred to as S2, is created when the atrioventricular valves open, causing blood to move

quickly from the atria to the ventricles. The base of the heart is the greatest place to hear S2, which is a higher-frequency sound [8].

2) *Transmission circuit*: The obtained analog signal, after going through the amplifier and filter, can be heard by ear-phones via a 3.5mm jack for auscultation purposes or being sent to an Arduino Uno R3 circuit. Arduino Uno R3 works as an ADC converting analog signals to digital signals ready for streaming to a remote device via Bluetooth.

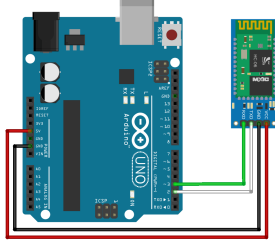


Fig. 5. Transmission circuit.

The transmission circuit is basically an Arduino UNO R3 connecting with an HC-06 Bluetooth module. The converted signal will be sent to the Excel program for storage via Bluetooth. To verify the operation of the transmission circuit, a known input is supplied into the board then the output data is transmitted to the laptop to be stored in Excel files. Stored data are plotted to calibrate the signal.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Simulation results

The electronic components listed in table I are selected from the calculations and specifications of the system.

TABLE I
CHOOSING COMPONENTS

Component	Symbol	Value
OpAmp	U_1, U_2	LTC6078
Capacitor	C_1, C_3	$47\mu F$
	C_2	$10\mu F$
	C_4	$300nF$
Resistor	R_2	$500k\Omega$
	R_1, R_3	$1M\Omega$
	R_5	$2k\Omega$

The circuit is then simulated by the TINA-TI simulation tool. The microphone is modeled by connecting a resistor in series with a capacitor. With $R_4 = 350k$, gain of AFE will be above 11 (21dB). An input signal of $200mV_{pp}$ is applied through the R_{mic} and the output signal is measured at V_{out} point. The simulation results are shown in Fig. 6 and Fig. 7.

From the transient simulation (Fig. 6), the output signal has an amplitude of $2.15V_{pp}$ offering a simulation gain of 10.75 or 20.63dB.

An AC analysis (Fig. 7) was carried out to obtain the frequency response of the system. It is noticed that AC gain equals 10.86 or 20.72dB at a bandwidth of 0 – 265Hz.

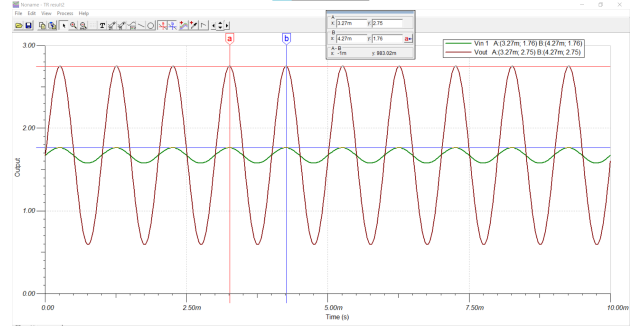


Fig. 6. Transient simulation results

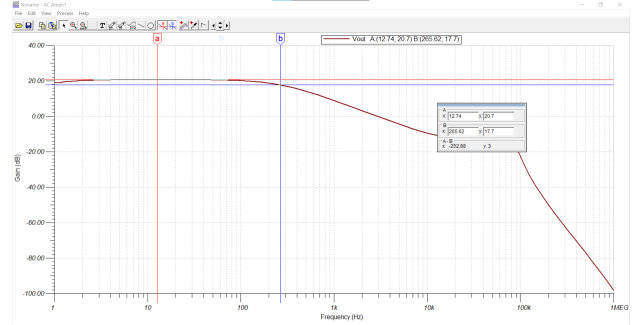


Fig. 7. AC analysis simulation results

B. Heart sound measurement

To determine the phases and shape of the signal, the measurements are set up with different steps. First, the chest piece is put on the chest of the subject (a volunteer). Second, the circuit is connected via Bluetooth to a laptop running the Excel program. Once in operation, the circuit continuously sends the output signals to the laptop for storage in Excel files. Each of these files contains 10 seconds of the heart sound. (Fig. 8) clearly confirms that the measured PCG signal has four different phases: S1, systole, S2, and diastole, as expected. The S1 peak is higher than the S2 peak.

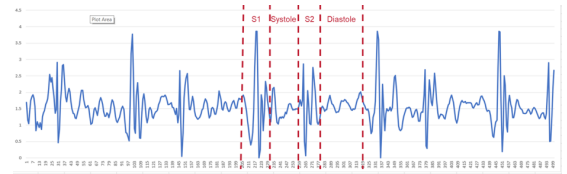


Fig. 8. Measured heart sound signal of one subject.

Table II shows the experiment results from 5 different subjects. For each subject, the values of the S1 peak and S2 peak are recorded and then averaged to detect the threshold for heartbeat counting.

C. Heart rate analysis

The calculation of heart rate can be summarized as follows:

- Storage of heart sound data in Excel file (10 seconds of sound for each file).

- Detection of meaningful signal peaks by using 'findpeak' function of Matlab and applying a threshold. The S1 peak, as the highest peak, is chosen to mark a heart cycle.
- Extraction of the time interval (Δt) between the current S1 peak and the next one (unit: second). Calculating the average of four consecutive time intervals.
- The heart rate can be calculated as follows:

$$HR = \frac{60(s)}{\Delta t(s)} \quad (3)$$

TABLE II
S1 AND S2 PEAKS RESULTS

	S1	S2	S3	S4	S5	AVG
S1 Peak(V)	3.40	3.23	3.35	3.61	3.20	3.36
S2 Peak(V)	2.75	2.63	2.72	2.71	2.59	2.68

To evaluate the heart rate counting algorithm, the heart rate results of the proposed electronic stethoscope are compared with commercially available pulse oximeters. The comparison is taken on 5 different subjects of different health conditions and genders, with ages ranging from 20 to 25 years old. We calculated the average heart rate measured by each product, then found the delta, which is determined as the difference between two HR AVG values.

The heart rate results (compared to commercial devices) (see table III) show that with a heart rate range from 75 to 85, our design nearly fits the commercial device results (delta range 0.7 to 1.0). However, with a heart rate lower or higher than that range, our system is not good. There are some reasons below that make this happen:

- The measurement results are strongly impacted by the environment (low noise environment required) and motion artifacts.
- The threshold level is fixed at a specific value while this value, in turn, varies depending on some factors such as health condition, gender, environment, or motion artifacts. It is obvious that this area needs further improvement by testing on a bigger group of subjects to identify the most efficient threshold level

TABLE III
HEART RATE COMPARING RESULTS

	Proposed product		Commercial product		Delta
	HR Range	AVG	HR Range	AVG	
S1	96, 84, 84, 72	84.0	84, 84, 83, 82	83.3	0.7
S2	82, 82, 79, 79	80.5	82, 81, 82, 81	81.5	1.0
S3	114, 114, 101, 113	110.5	89, 90, 89, 91	89.8	20.7
S4	66, 66, 66, 60	64.5	69, 70, 69, 72	70.0	5.5
S5	66, 60, 60, 60	61.5	64, 64, 64, 65	64.3	2.8

IV. CONCLUSION

This study described the development of a real-time and low-cost electronic stethoscope for the auscultation of heart sounds. For use in medical diagnosis and auscultation learning, the stethoscope produces clear sound recordings. The PCG was

used to determine the heart rate, and the results were compared with a commercial device. The design will be improved in the future by optimizing it to increase performance.

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